

Portfolio Risk Monitor and Multi-Asset Operations Layer

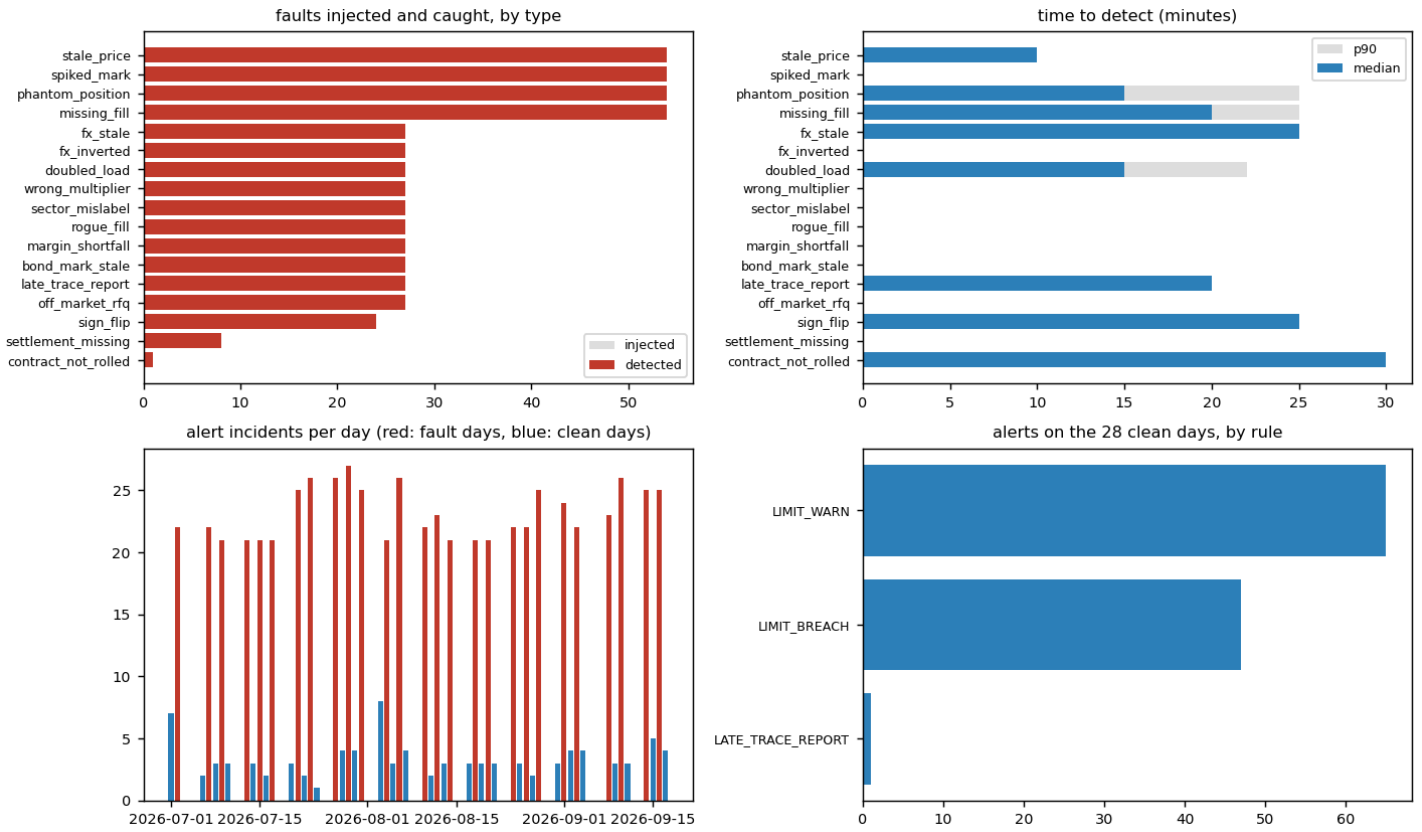
Question. What does a systematic multi-asset book look like intraday, how fast is a limit breach, a bad feed, a position break or a margin call caught, and what changes when the book holds futures, FX forwards and corporate bonds rather than US equities alone? This layer extends execution-ops (its equity book, calendars and fault-injection method) with positions-to-exposures, limits, intraday P&L attribution and margin on a virtual clock, futures rolls and first-notice handling with a SPAN-style margin, FX forwards with value dates, settlement instructions and CLS-style netting, and a credit RFQ post-trade against evaluated marks and TRACE-style prints; the monitor is fault-injected and scored like the order monitor.

Method. Python package (xrisk) on DuckDB. The book: execution-ops' three equity strategies on \$2bn, an index-futures hedge, a Treasury-futures steepener, a commodity and yen macro book, a corporate-bond book hedged with Treasury futures, and an FX-forward hedge of the non-dollar equity exposure. A risk model estimated from the daily data (regional market, sector and style factors for equities; own return factors for futures and FX; par-yield and credit-spread factors with measured DV01s for Treasury futures and bonds) gives exposures, parametric and historical VaR and the attribution. Every daily mark is real; the intraday path between two marks is a Brownian bridge. Fault days alternate with clean days.

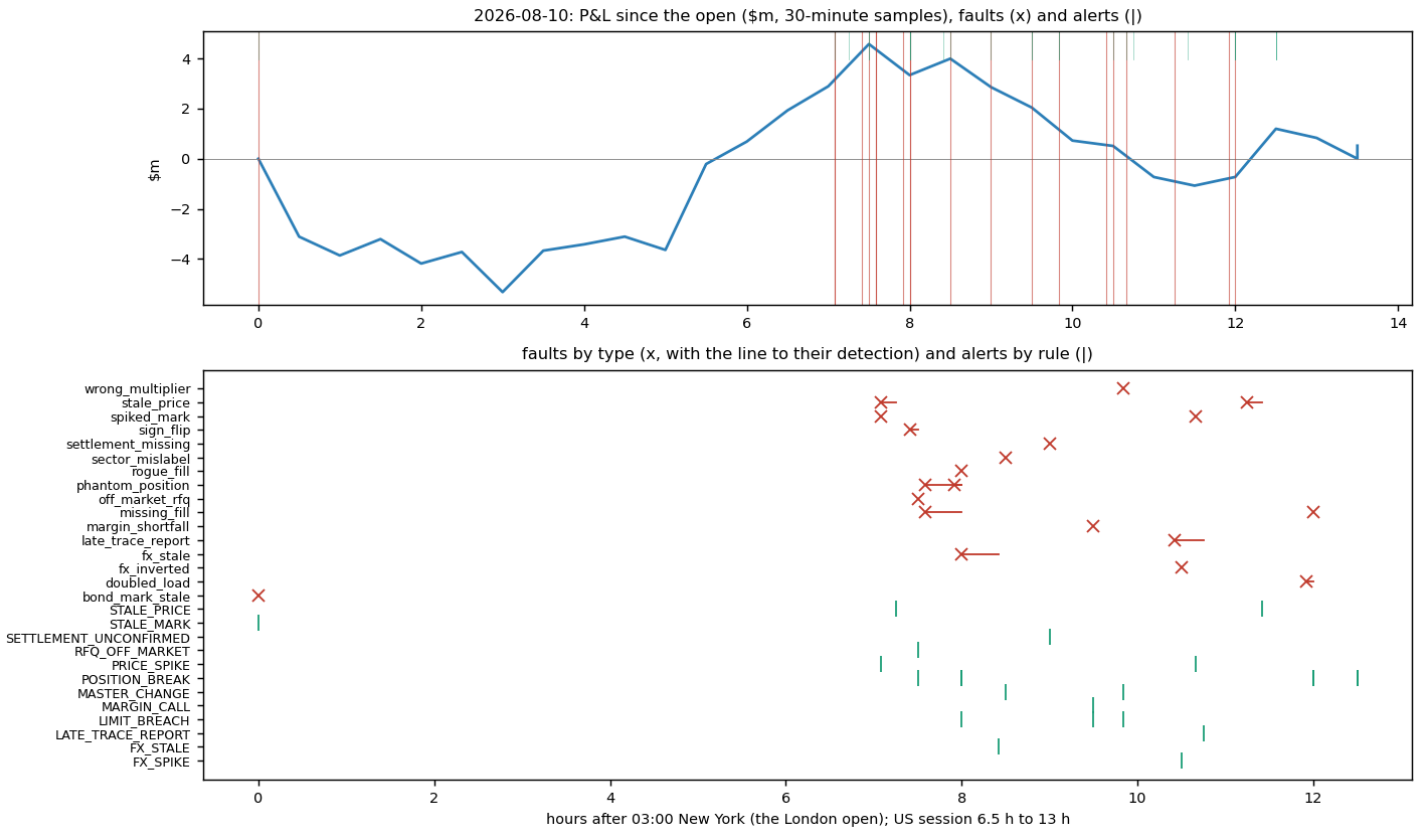
Caveats. Fills, the prime broker, the dealers and TRACE prints are simulated; prices, settlements, FX fixes, rates, bond marks (one date), calendars and their inconsistencies are real. Yahoo lists futures contracts only while they trade, so earlier front months come from its continuous series where the check shows it equals the front. The SPAN credit rates and the margin haircuts are this model's parameters.



The book: gross by strategy and asset class, exposures over the run, VaR against the daily loss, net sector exposure.

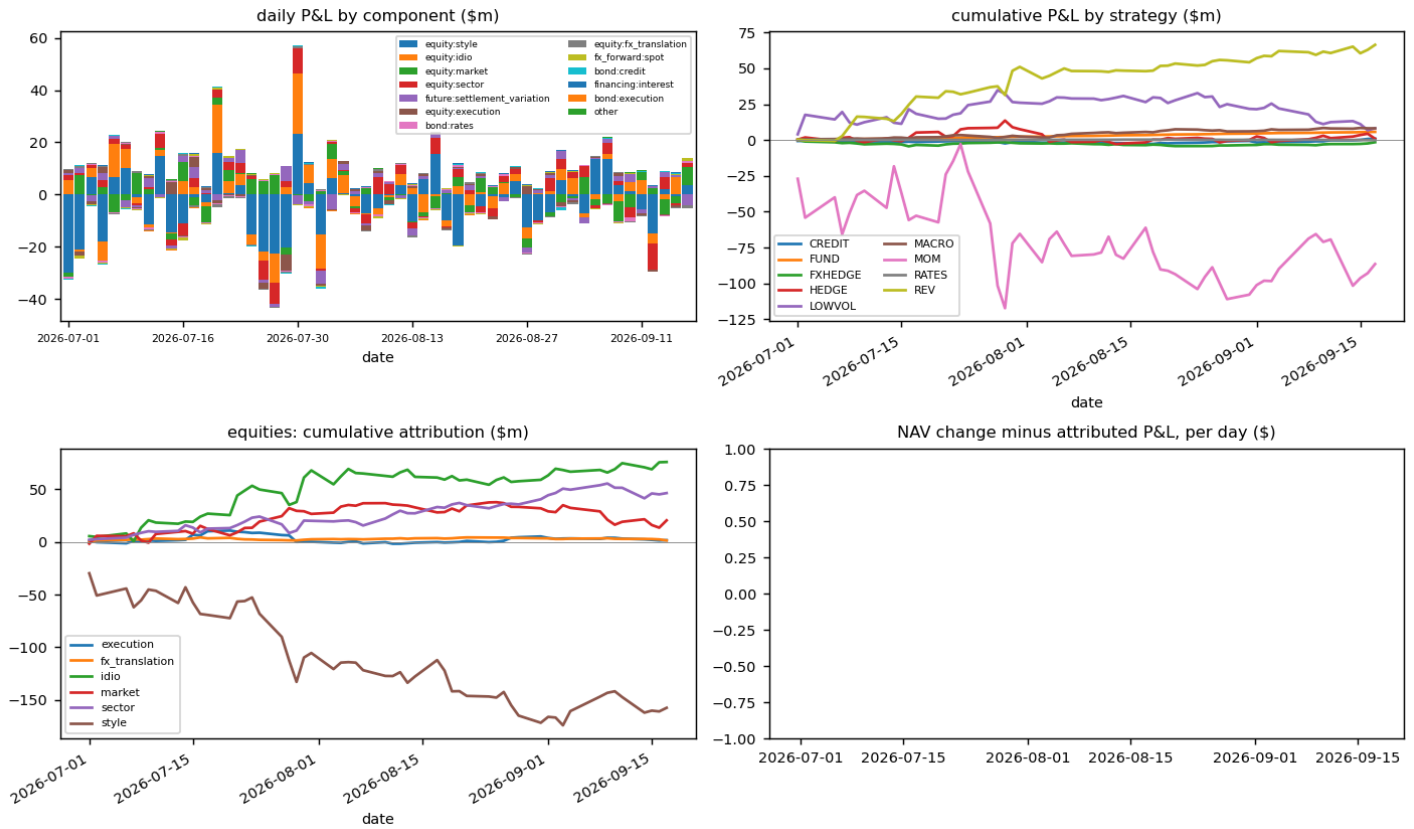


Fault injection: faults injected and caught by type, time to detect, incidents per day on fault and clean days, alerts on clean days by rule.

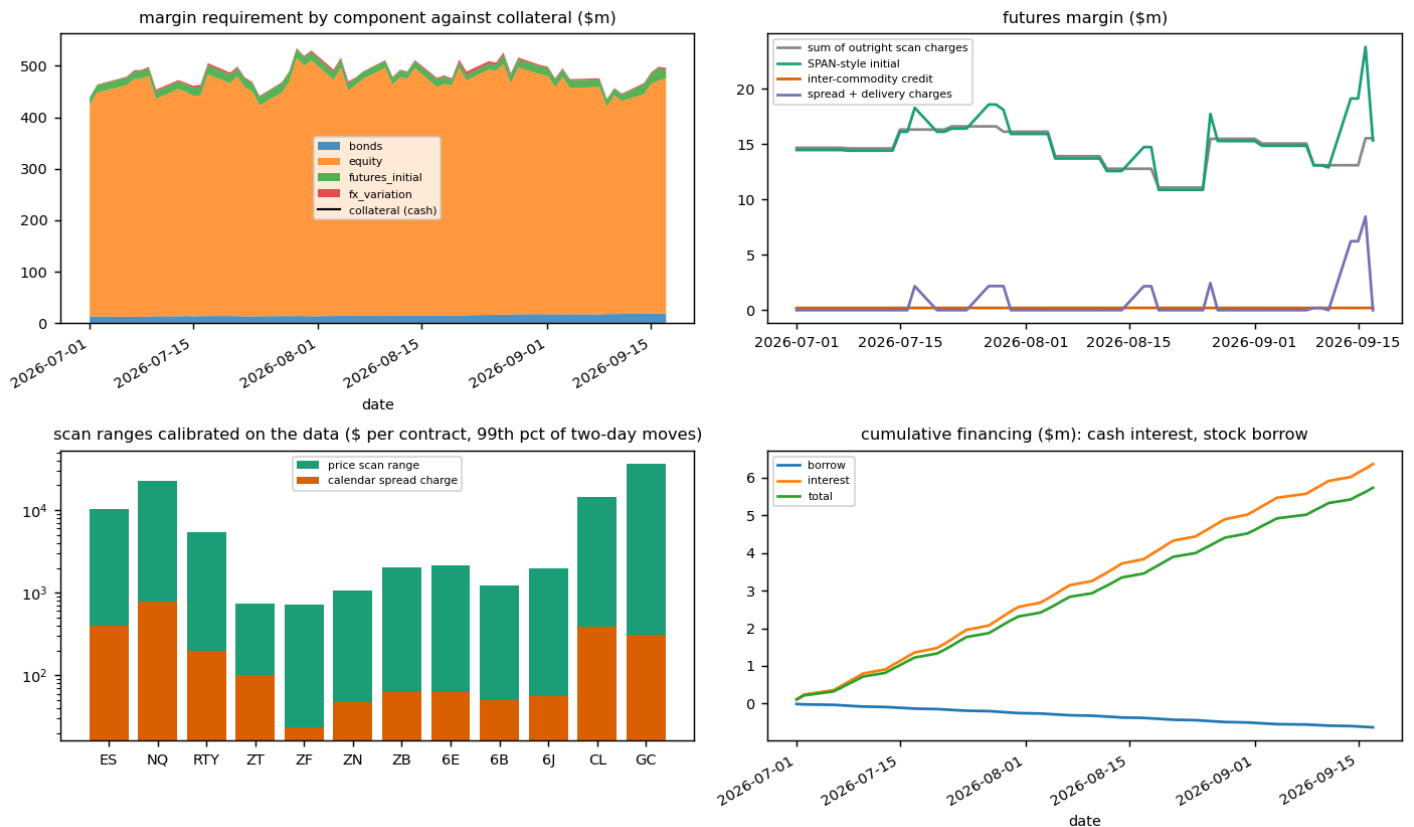


One fault day: P&L since the open with the faults and the alerts, then faults by type with the line to their detection and alerts by rule.

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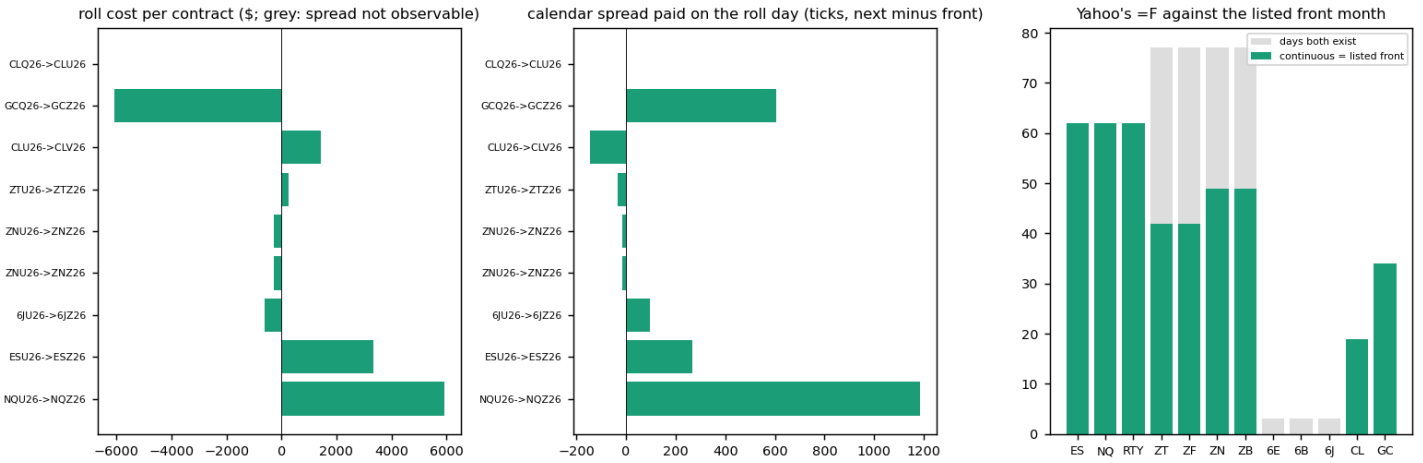


P&L attribution: daily by component, cumulative by strategy, the equity components, and the identity gap per day.

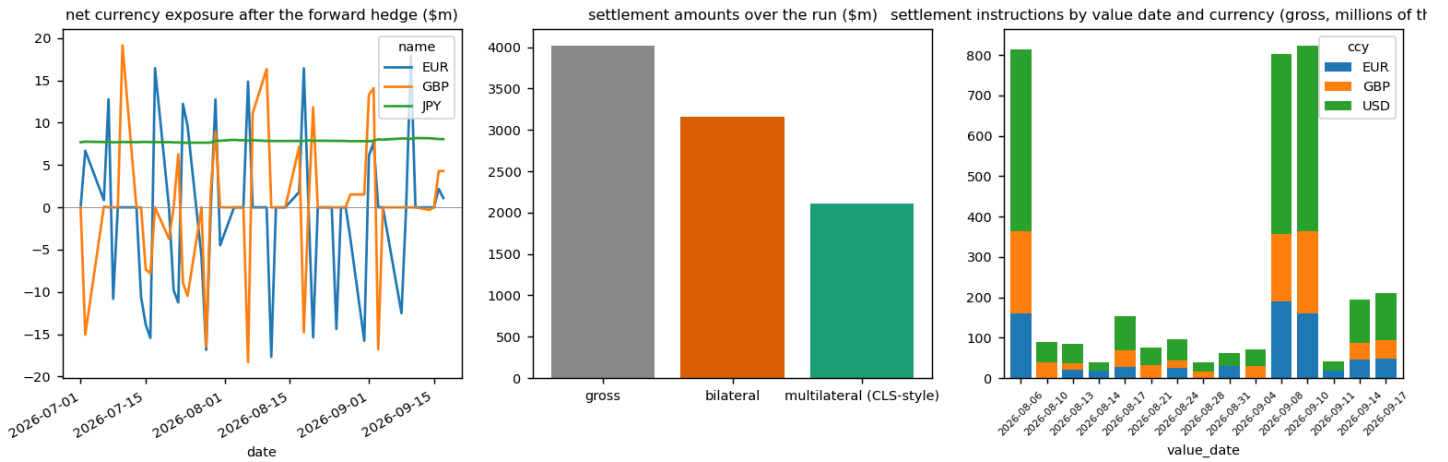


Margin: requirement against collateral, the SPAN-style futures margin against the sum of outrights, the calibrated scan ranges, financing.

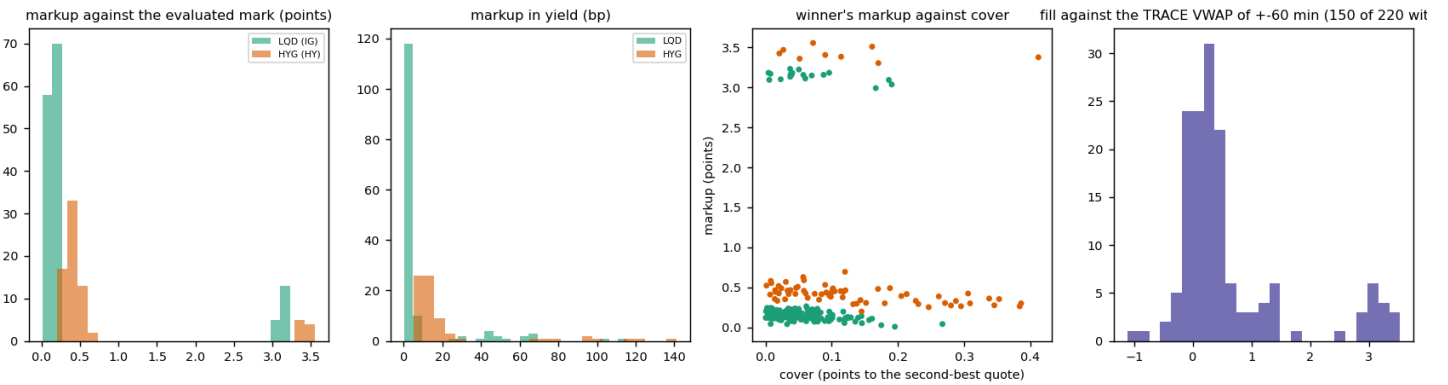
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Futures rolls: cost per contract and calendar spread per roll, and the continuous-versus-front check on the data.

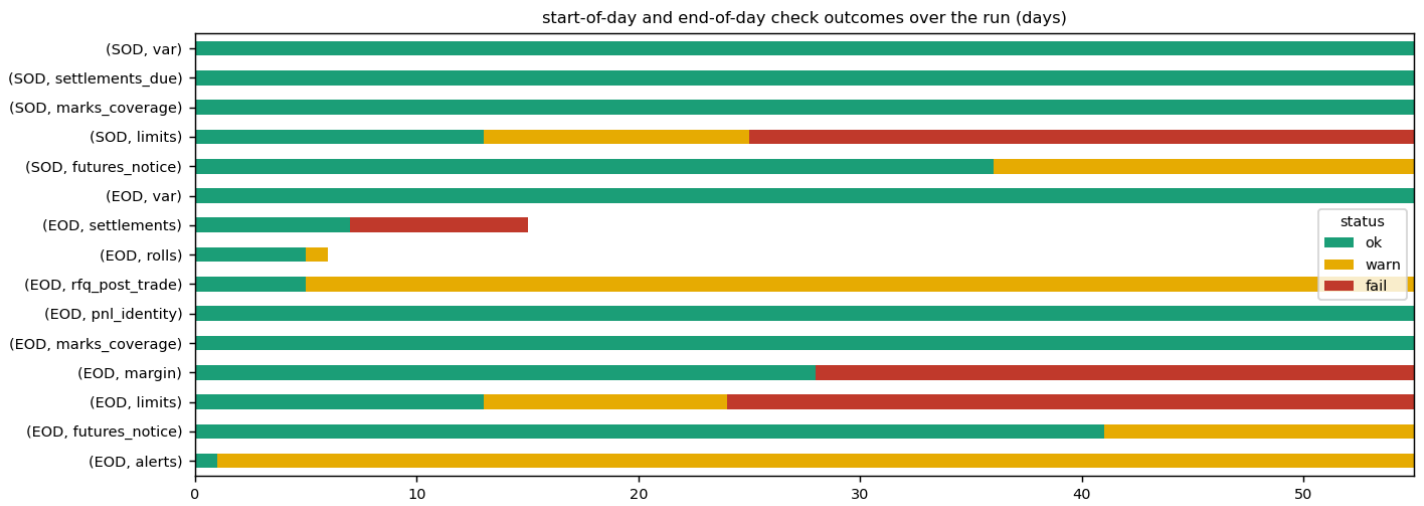


FX: net currency exposure after the hedge, settlement amounts gross versus bilateral versus multilateral, instructions by value date.



Credit RFQs: markup in points and in yield by bucket, markup against cover, fills against the TRACE VWAP.

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Start-of-day and end-of-day check outcomes over the run.

Generated by scripts/summarize.py from results/run.json and data/derived/xrisk.duckdb.
Window 2026-07-01 to 2026-09-17: 55 trading days, 27 with faults injected and 28 clean, alternating. NAV \$2bn at the start, \$2.002bn at the end. Runtime 57 min on the build machine. Calendars from xops (sibling).

Risk monitor under fault injection

519 of 519 faults detected over 27 fault days (100.0 %), median time to detect 0 s, p90 25 min. On the 28 clean days the monitor raised 4.0 alerts a day (3.4 distinct incidents).

						fault
stale_price	54	54	10 min	10 min	STALE_PRICE (54)	JUMP (1)
spiked_mark	54	54	0 s	0 s	PRICE_SPIKE (54)	
phantom_position	54	54	15 min	25 min	POSITION_BREAK (54)	
missing_fill	54	54	20 min	25 min	POSITION_BREAK (54)	
fx_stale	27	27	25 min	25 min	FX_STALE (27)	
fx_inverted	27	27	0 s	0 s	FX_SPIKE (27)	
doubled_load	27	27	15 min	22 min	POSITION_BREAK (27)	
wrong_multiplier	27	27	0 s	0 s	MASTER_CHANGE (27)	
sector_mislabel	27	27	0 s	0 s	MASTER_CHANGE (27)	
rogue_fill	27	27	0 s	0 s	LIMIT_BREACH (26), PNL_JUMP (1)	
margin_shortfall	27	27	0 s	0 s	MARGIN_CALL (27)	
bond_mark_stale	27	27	0 s	0 s	STALE_MARK (27)	
late_trace_report	27	27	20 min	20 min	LATE_TRACE_REPORT (27)	
off_market_rfq	27	27	0 s	0 s	RFQ_OFF_MARKET (27)	
sign_flip	24	24	25 min	25 min	POSITION_BREAK (24)	
settlement_missing	8	8	0 s	0 s	SETTLEMENT_UNCONFIRMED (8)	(8)
contract_not_rolled	1	1	30 min	30 min	ROLL_OVERDUE (1)	

Alerts on clean days by rule:

		rule
LIMIT_WARN	65	
LIMIT_BREACH	47	
LATE_TRACE_REPORT	1	

The book, its exposures and limits

strategy	asset class	positions	gross \$m	net \$m
CREDIT	bond	60	276.5	148.5
CREDIT	future	1	193.7	-193.7
FXHEDGE	fx_forward	39	1,887.0	-148.6
HEDGE	future	2	288.0	-288.0
LOWVOL	equity	60	595.5	595.5
MACRO	future	3	49.7	49.7
MOM	equity	112	1,940.1	21.5
RATES	future	2	119.2	67.6
REV	equity	74	402.9	2.8

Averages at the close over the run: gross \$4.21bn (2.11 x NAV), net \$339m, beta-dollars \$397m after the index-futures hedge, DV01 \$536/bp, parametric 99 % one-day VaR \$23.0m (1.15 % of NAV), historical \$26.2m. Daily P&L: mean \$+0.04m, sd \$16.19m, worst \$-35.11m on 2026-07-28; the 99 % VaR was exceeded on 5 of 55 days (parametric) and 5 (historical).
Limit evaluations at the open and the close (rows are limit x scope x day):

				limit
beta_dollars	89	21	0	
cs01	220	0	0	
currency_net	328	0	1	
dv01	109	0	1	
dv01_strategy	218	0	1	
futures_root	763	0	0	
gross	110	0	0	
intraday_loss	51	1	3	
liquidity	84	8	18	
margin	55	0	0	

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net_equity	110	0	0
sector_net	1849	19	2
single_name	1044	9	47
var	110	0	0

Most frequent breaches: liquidity:days_to_liquidate on 11 days (worst 392 %); single_name:TJX on 4 days (worst 112 %); single_name:CSX on 4 days (worst 112 %); intraday_loss:all on 3 days (worst 116 %); sector_net:Financial Services on 2 days (worst 114 %); single_name:MAR on 2 days (worst 109 %); single_name:UNP on 2 days (worst 113 %); single_name:XRX on 2 days (worst 105 %)

P&L attribution

Total P&L \$+2.4m over the run; the attribution reconciles to the NAV change on every day (largest gap 0.0000 USD).

		asset class
bond:carry	+2.24	
bond:credit	+1.00	
bond:execution	-3.35	
bond:idio	+0.54	
bond:rates	-5.05	
equity:execution	+1.61	
equity:fx_translation	+1.40	
equity:idio	+75.91	
equity:market	+20.46	
equity:sector	+46.37	
equity:style	-157.58	
financing:borrow	-0.63	
financing:interest	+6.36	
future:execution	-0.11	
future:settlement_variation	+14.71	
fx_forward:execution	-0.82	
fx_forward:forward_points	-0.36	
fx_forward:settled	+1.12	
fx_forward:spot	-1.44	
strategy	\$m	
CREDIT	+0.95	
FUND	+5.73	
FXHEDGE	-1.50	
HEDGE	+0.91	
LOWVOL	+8.21	
MACRO	+8.19	
MOM	-86.46	
RATES	-0.07	
REV	+66.42	

Margin and financing

Requirement \$487m on average against cash collateral, utilisation 40 % on average and 45 % at worst (clean days; the margin-shortfall fault raises it above 100 % on fault days). Futures initial margin \$15.0m SPAN-style against \$14.5m as the sum of outright scan charges. Financing over the run \$+5.73m (cash interest less stock borrow).

			root
ES	10,325	400	
NQ	22,480	790	
RTY	5,440	200	
ZT	734	102	
ZF	719	23	
ZN	1,062	47	
ZB	2,031	62	
6E	2,156	62	
6B	1,212	50	
6J	1,950	56	

CL	14,440	390
GC	36,280	310

Futures rolls

date	roll	lots	front	next	spread (ticks)	roll P&L \$/contract (pos	spread observed
2026-07-20 00:00:00	CLQ26->CLU26	+150	83.2300	83.2300	+0.0	-23	no (next month not liste
2026-07-30 00:00:00	GCQ26->GCZ26	+60	4100.1001	4160.6001	+605.0	-6,073	yes
2026-08-19 00:00:00	CLU26->CLV26	+150	85.8300	84.3900	-144.0	+1,417	yes
2026-08-27 00:00:00	ZTU26->ZTZ26	+462	102.9766	102.8398	-35.0	+256	yes
2026-08-27 00:00:00	ZNU26->ZNZ26	-242	108.5781	108.3281	-16.0	-283	yes
2026-08-27 00:00:00	ZNU26->ZNZ26	-1743	108.5781	108.3281	-16.0	-283	yes
2026-09-11 00:00:00	6JU26->6JZ26	+100	0.0065	0.0066	+96.0	-616	yes
2026-09-17 00:00:00	ESU26->ESZ26	-573	7640.0000	7707.2500	+269.0	+3,334	yes
2026-09-17 00:00:00	NQU26->NQZ26	-113	29446.7500	29743.0000	+1185.0	+5,912	yes

root	rolls	lots	observed	mean spread (ticks)	mean roll P&L \$/contract	total \$
6J	1	100	1	+96.0	-616	-61,570
CL	2	300	1	-144.0	+1,417	+209,100
ES	1	573	1	+269.0	+3,334	+1,910,668
GC	1	60	1	+605.0	-6,073	-364,392
NQ	1	113	1	+1185.0	+5,912	+668,056
ZN	2	1985	2	-16.0	-283	-562,251
ZT	1	462	1	-35.0	+256	+118,185

Yahoo's continuous series against the listed front month (where both exist): ES 62/62; NQ 62/62; RTY 62/62; ZT 42/77; ZF 42/77; ZN 49/77; ZB 49/77; 6E 0/3; 6B 0/3; 6J 0/3; CL 19/19; GC 34/34.

FX forwards and settlement

75 forward trades, 70 settlement instructions on 15 value dates; gross \$4,013m, bilateral payment netting \$3,160m, multilateral (CLS-style) \$2,107m: a 47 % reduction; 8 unconfirmed at the cutoff (all on fault days, the settlement_missing fault).

						value da
2026-08-06	EURUSD	EUR	DLR2	66,434,000	True	
2026-08-06	EURUSD	EUR	DLR4	93,754,000	True	
2026-08-06	EURUSD	USD	DLR2	-75,534,166	False	
2026-08-06	EURUSD	USD	DLR4	-106,940,992	True	
2026-08-06	GBPUSD	GBP	DLR2	-148,400,000	True	
2026-08-06	GBPUSD	GBP	DLR2	-54,605,000	True	
2026-08-06	GBPUSD	USD	DLR2	195,922,651	True	
2026-08-06	GBPUSD	USD	DLR2	72,284,304	True	
2026-08-10	GBPUSD	GBP	DLR1	18,183,000	True	
2026-08-10	GBPUSD	GBP	DLR2	-19,955,000	True	
2026-08-10	GBPUSD	USD	DLR1	-24,356,103	False	
2026-08-10	GBPUSD	USD	DLR2	26,633,732	True	

Credit RFQ post-trade

bucket	trades	mean markup (pts)	median	mean markup (bp yield)	mean cover (pts)	mean dev. vs TRACE VWAP	warn prints	flagged
HYG (HY)	74	0.778	0.428	22.9	0.131	0.696	42	41
LQD (IG)	146	0.520	0.163	9.0	0.057	0.587	108	63

Flags: {'': 116, 'no_trace_prints': 49, 'off_market_vs_trace,off_market_vs_mark': 17, 'late_trace_report': 16, 'no_trace_prints,late_trace_report': 12, 'no_trace_prints,off_market_vs_mark': 9, 'off_market_vs_mark': 1}. Winning dealers: {'DLR1': 113, 'DLR2': 58, 'DLR3': 37, 'DLR4': 12}. Late TRACE reports: 28 (the late_trace_report fault).

Checks

phase	check	ok	warn	fail
EOD	alerts	1	54	0
EOD	futures_notice	41	14	0
EOD	limits	13	11	31
EOD	margin	28	0	27

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EOD	marks_coverage	55	0	0
EOD	pnl_identity	55	0	0
EOD	rfq_post_trade	5	50	0
EOD	rolls	5	1	0
EOD	settlements	7	0	8
EOD	var	55	0	0
SOD	futures_notice	36	19	0
SOD	limits	13	12	30
SOD	marks_coverage	55	0	0
SOD	settlements_due	55	0	0
SOD	var	55	0	0